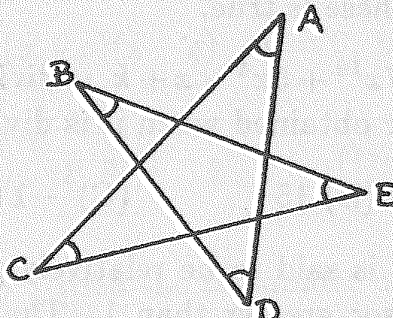


ITMA 94  
SCREENING TEST - JUNIOR LEVEL - 1994

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1. If  $\log_x 4 = 0.4$ , then the value of  $x$  is  
(A) 4                      (B) 16                      (C) 1                      (D) 32                      (E) None of these.
2. A circular swimming pool is surrounded by a concrete walk 4 feet wide. If the area of the walk is  $\frac{11}{25}$  that of the pool, then the radius of the pool is  
(A) 20ft                      (B) 8ft                      (C) 30ft                      (D) 16ft                      (E) None of these.
3. If  $x$  and  $y$  are positive integers such that  $1994 = (x + y)^2 + 3x + y$ , then  $y$  is equal to  
(A) 19                      (B) 37                      (C) 30                      (D) 94                      (E) None of these.
4. In the star-shaped figure shown here, the sum of the angles marked at the corners  $A, B, C, D, E$  is  
(A)  $90^\circ$   
(B)  $135^\circ$   
(C)  $180^\circ$   
(D)  $225^\circ$   
(E) None of these.
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5. If  $a, b, c$  are positive real numbers such that  $\frac{a}{2b-c} = \frac{2b}{3a+c} = \frac{a}{b}$ , then  $\frac{a}{b}$  is equal to  
(A)  $\frac{1}{2}$                       (B)  $\frac{2}{3}$                       (C)  $\frac{3}{4}$                       (D)  $\frac{5}{4}$                       (E) None of these.
6. In a box there are coins and rings, which are either silver or gold. If 30% of the objects in the box are coins and 40% of the rings are silver, what percentage of the objects in the box are gold rings?  
(A) 60%                      (B) 30%                      (C) 42%                      (D) 70%                      (E) None of these.

7. In triangle  $ABC$ ,  $\angle BAC = 30^\circ$ . Then
- (A)  $BC$  is the inradius of the triangle  $ABC$ .
  - (B)  $BC$  is the circumradius of the triangle  $ABC$ .
  - (C)  $BC$  is the side of the square having the same area as triangle  $ABC$ .
  - (D)  $BC$  is the side of an equilateral triangle which has the same area as triangle  $ABC$ .
  - (E) None of these is true.
8. If 10 balls are placed in 4 boxes at random so that no box is empty, then
- (A) exactly one box has exactly three balls
  - (B) at least one box has exactly three balls
  - (C) exactly one box has at least three balls
  - (D) at least one box has at least three balls
  - (E) none of these is true.
9. If  $a, b, c$  are positive integers and two of them are odd, then  $3^a + (b-1)^2(c-1)^3$  is
- (A) even for all such choices.
  - (B) odd for all such choices.
  - (C) even if  $a$  is even and odd if  $a$  is odd.
  - (D) even if  $a$  is odd and odd if  $a$  is even.
  - (E) None of these is true.
10. When  $x^{94} - 7x^{19} + 5x^4 - x + k$  is divided by  $x - 1$ , the remainder is  $-1$ . Then the remainder obtained when it is divided by  $x + 1$  is
- (A) 0
  - (B) 15
  - (C)  $-1$
  - (D) 10
  - (E) None of these.
11. An integer  $a$  is said to be relatively prime to an integer  $b$  if  $a$  and  $b$  have no common divisor greater than 1. The number of positive integers less than 175 and relatively prime to both 5 and 7 is
- (A) 140
  - (B) 125
  - (C) 120
  - (D) 100
  - (E) None of these.
12. A circle has radius 2 cm. A square is inscribed in the circle. The area of the square is
- (A) 2 sq.cm
  - (B) 4 sq.cm
  - (C)  $2\sqrt{2}$  sq.cm
  - (D) 8 sq.cm
  - (E) None of these.
13. In a triangle  $ABC$ ,  $AB = AC$ ,  $BC = 6$  cm and  $\angle A = 90^\circ$ . If  $G$  is the centroid of triangle  $ABC$ , then  $AG$  is equal to



4. A triangle has sides 8,17,15. The area of the triangle is  
 (A) 50 (B) 68 (C) 40 (D) 60 (E) None of these.
5. If  $a = (\sqrt{3} + \sqrt{2})^{-3}$  and  $b = (\sqrt{3} - \sqrt{2})^{-3}$ , then  $(a + 1)^{-1} + (b + 1)^{-1} = \dots$   
 (A)  $50\sqrt{3}$  (B)  $48\sqrt{2}$  (C) 1 (D) 5 (E) None of these.
6. If  $\sqrt{x^2 + 9} + \sqrt{x^2 - 3} = 6$ , then  $\sqrt{x^2 + 9} - \sqrt{x^2 - 3} = \dots$   
 (A) 1 (B) 3 (C) 5 (D) 2 (E) None of these.
7. The sides of a rhombus are two radii and two chords of a circle. If the radius of the circle is 10 cm, then the area of the rhombus is  
 (A)  $50\sqrt{3}$  sq.cm (B) 100 sq.cm (C)  $100\sqrt{3}$  sq.cm (D)  $\frac{100}{\sqrt{3}}$  sq.cm  
 (E) None of these.
8. A right circular cylinder open at both ends has base radius  $5/\pi$  cm and height 12cm. If  $A$  and  $B$  are diametrically opposite points lying on the top and bottom rims, then the shortest distance between  $A$  and  $B$  along the surface of the cylinder is  
 (A) 13 cm (B) 17 cm (C)  $12 + \frac{10}{\pi}$  cm  
 (D)  $\sqrt{12^2 + (\frac{10}{\pi})^2}$  cm (E) None of these.
9. If  $m$  and  $n$  are the roots of the equation  $(x + p)(x + q) - k = 0$ , then the roots of  $(x - m)(x - n) + k = 0$  are  
 (A)  $p$  and  $q$  (B)  $\frac{1}{p}$  and  $\frac{1}{q}$  (C)  $-p$  and  $-q$  (D)  $mp$  and  $nq$   
 (E) None of these.
10. If  $f(x) = x^3 + ax + b$  is divisible by  $(x - 1)^2$ , then the remainder obtained when  $f(x)$  is divided by  $x + 2$  is  
 (A) 1 (B) -2 (C) 3 (D) -1 (E) None of these.
11. In a triangle  $ABC$ , the midpoint of the side  $BC$  is  $D$  and the centroid is  $G$ . Then the ratio of the area of the triangle  $BGD$  to the area of the triangle  $ABC$  is  
 (A)  $\frac{1}{2}$  (B)  $\frac{1}{4}$  (C)  $\frac{1}{3}$  (D)  $\frac{1}{9}$  (E) None of these.

22. Two circles having radii 5 and 3 touch each other internally. If the perpendicular bisector of the segment joining their centres meets the outer circle in  $P$  and  $Q$ , then  $PQ$  is equal to
- (A)  $4\sqrt{6}$  (B) 10 (C) 6 (D)  $2\sqrt{26}$  (E) None of these.
23. When  $3^{1994} + 2$  is divided by 11 the remainder is
- (A) 3 (B) 4 (C) 6 (D) 0 (E) 1.
24. The area of a circle circumscribed about a regular hexagon is  $2\pi$ . The area of the hexagon is
- (A) 6 (B)  $3\sqrt{3}$  (C)  $\frac{\sqrt{3}}{2}$  (D)  $6\sqrt{3}$  (E)  $3\sqrt{6}$
25. The angle between the hands of a clock at 1.15 is
- (A)  $60^\circ$  (B)  $50^\circ$  (C)  $52\frac{1}{2}^\circ$  (D)  $55^\circ$  (E)  $56\frac{1}{4}^\circ$ .
26. If 12 divides  $ab313ab$ , the smallest value of  $a + b$  is
- (A) 2 (B) 4 (C) 6 (D) 7 (E) 10.
27.  $a, b, c$  are positive integers such that  $\frac{a}{3} = \frac{b}{4} = \frac{c}{5}$  and  $abc = 1620$ . The value of  $b$  is
- (A) 8 (B) 10 (C) 12 (D) 6 (E) None of these.
28. The last digit of  $7^{111}$  is
- (A) 7 (B) 3 (C) 1 (D) 9 (E) None of these.
29. If  $abcd = (abc) \cdot (bd)$  where  $a, b, c, d$  stand for digits, the value of  $b$  is
- (A) 0 (B) 1 (C) 2 (D) 5 (E) 7
30. The last 2 digits of  $25^{63} \times 63^{25}$  are
- (A) 25 (B) 75 (C) 55 (D) 45 (E) 35.